

SIGN PERIODICITY FOR INTEGER POWERS OF THE INFINITE BORWEIN PRODUCT: A PROOF OF WANG'S CONJECTURE

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ABSTRACT. For integers $t \geq 2$ and $m \geq 1$ let $c_t^{(m)}(n)$ denote the coefficients of $(q; q)_\infty^m / (q^t; q^t)_\infty^m$. L. Wang conjectured that the sign of $c_t^{(m)}(n)$ is eventually periodic in n , with least period divisible by t . He proved this when $m(t-1) \leq 24$, for $t = 2$, and for prime t with $m \in \{1, 3\}$. We prove the conjecture in general.

The argument has four parts:

- a Fourier argument on $\mathbb{Z}/t\mathbb{Z}$, which shows that any eventual period is divisible by t ;
- a uniqueness principle for Rademacher–Zuckerman expansions, which lets us work with groups of terms of equal exponential growth rather than with individual terms;
- a Galois scaling law for these groups, obtained from an explicit Chinese-remainder factorisation of the amplitudes: scaling a modulus by a prime multiplies a group amplitude by a nonzero Kloosterman sum and applies a Galois automorphism;
- a finiteness argument showing that on every residue class the sign is decided by one of finitely many “rigid” groups.

For prime t we also give a direct proof through Kloosterman and Salié sums of prime-power modulus, and we determine the sign pattern explicitly when m is even.

1. INTRODUCTION

1.1. The conjecture. For integers $t \geq 2$ and $m \geq 1$ write

$$(1) \quad G_t(q)^m = \frac{(q; q)_\infty^m}{(q^t; q^t)_\infty^m} = \sum_{n \geq 0} c(n) q^n, \quad c(n) = c_t^{(m)}(n) \in \mathbb{Z},$$

where $(q; q)_\infty = \prod_{j \geq 1} (1 - q^j)$. Andrews [1] proved that $c_p^{(1)}(n) c_p^{(1)}(n+p) \geq 0$ for primes p . Schlosser and Zhou [9] studied real powers of G_p . Wang [10] proved that $\text{sgn } c(n)$ is eventually periodic whenever $m(t-1) \leq 24$, and determined the sign patterns in that range. Beyond this range he stated the following.

Definition 1.1. A real sequence $(a(n))$ is *ultimately periodic in sign* (UPS) if there is $L \geq 1$ with $\text{sgn } a(n+L) = \text{sgn } a(n)$ for all sufficiently large n . The least such L is the *least period of sign*.

Conjecture (Wang [10, Conj. 1.6]). For all positive integers t and m , the sequence $(c_t^{(m)}(n))_{n \geq 0}$ is UPS, and its least period of sign is divisible by t .

Wang proved the conjecture for $m(t-1) \leq 24$, for $t = 2$ and all m , and for prime t with $m \in \{1, 3\}$.

Theorem 1.2. *Wang’s conjecture holds for all integers $t \geq 2$ and $m \geq 1$.*

For prime level there is a more explicit statement. Let

$$\alpha(r) = \sum_{\substack{h \bmod t \\ \gcd(h,t)=1}} e^{-\pi i m s(h,t)} e\left(-\frac{hr}{t}\right),$$

where $s(h, t)$ is the Dedekind sum and $e(x) = e^{2\pi i x}$. This is Wang’s leading amplitude [10, (1.21)].

Theorem 1.3. *Let $t = p$ be prime.*

- (a) *If $p \geq 5$ and m is even, or if $p = 3$ and $m \not\equiv 3 \pmod{6}$, then $\alpha(r) \neq 0$ for every residue r . Consequently $\operatorname{sgn} c(n) = \operatorname{sgn} \alpha(n \bmod p)$ for all large n , and the least period of sign is exactly p .*
- (b) *If $p \geq 5$ and m is odd, then $\alpha(r) = 0$ exactly when $\chi_p((\mu - r)\mu) = -1$, or when $r \equiv \mu \equiv 0 \pmod{p}$. Here χ_p is the Legendre symbol and $\mu \equiv -m\overline{24} \pmod{p}$. On each such class the sign is eventually constant (possibly 0).*

1.2. Strategy. Put $\mu = m(t-1)/24$ and $F(\tau) = \eta(\tau)^m / \eta(t\tau)^m = q^{-\mu} \sum c(n)q^n$. This is a modular function of weight 0 on $\Gamma_0(t)$ with a multiplier of finite order. The Rademacher–Zuckerman exact formula expresses $c(n)$ as an absolutely convergent series of terms

$$A_T(n) B_T(n).$$

The amplitude A_T is a generalized Kloosterman sum, periodic in n . The Bessel factor B_T grows like $\exp(4\pi\beta_T\sqrt{n})$ for a growth rate $\beta_T > 0$.

On a residue class, the sign of $c(n)$ is decided by the fastest-growing term that does not vanish. There are two difficulties:

- (i) terms of different cusp types can have exactly the same growth rate and cancel;
- (ii) infinitely many terms exist, so a priori the deciding term could have an unbounded denominator, which would destroy periodicity.

We handle (i) by working with groups of terms of equal growth rate, whose amplitudes are canonical by a uniqueness principle (Lemma 2.4). We handle (ii) with a Galois scaling law for groups (Proposition 5.3): a non-rigid group can be nonzero only where a faster rigid group is nonzero. This is combined with a finiteness argument (Section 6). Divisibility of the period by t follows from a Fourier argument (Section 3).

Notation.

- $e(x) = e^{2\pi ix}$.
- $\bar{a}$ denotes an inverse of a modulo a modulus that will be clear from context.
- χ_p is the Legendre symbol, and $(\cdot)$ is the Jacobi or Kronecker symbol.
- For a set of primes S , an S -unit is a rational number whose numerator and denominator have all their prime factors in S .

2. EXACT FORMULAS AND THE UNIQUENESS PRINCIPLE

2.1. Two modular realisations. Let $F(\tau) = \eta(\tau)^m \eta(t\tau)^{-m}$ and $N = 576t$.

Lemma 2.1. *$F^*(\tau) := F(24\tau) = \eta(24\tau)^m \eta(24t\tau)^{-m}$ is a weakly holomorphic modular function on $\Gamma_0(N)$ with the real character $\chi(d) = (\frac{t-m}{d})$. Moreover*

$$F^*(\tau) = \sum_{n \geq 0} c(n) q^{24n - m(t-1)}.$$

Proof. The exponents are $r_{24} = m$ and $r_{24t} = -m$. We have $\sum_{\delta} \delta r_{\delta} = 24m(1-t) \equiv 0 \pmod{24}$ and $\sum_{\delta} (N/\delta) r_{\delta} = 24tm - 24m \equiv 0 \pmod{24}$. The weight is $\frac{1}{2} \sum r_{\delta} = 0$. By the Gordon–Hughes–Newman theorem [8, Thm. 1.64], F^* is modular on $\Gamma_0(N)$ with character $\chi(d) = (\frac{(-1)^0 s}{d})$, where $s = \prod \delta^{r_{\delta}} = t^{-m}$. This character takes values ± 1 . Finally, F^* is holomorphic on $\mathfrak{H}$, and at every cusp it has an expansion with finitely many negative powers. $\square$

Theorem 2.2 (Zuckerman [11], Bringmann–Ono [3]). *Let f be a weakly holomorphic modular function on $\Gamma_0(N)$ with character. Then for every $x > 0$ its x -th Fourier coefficient at ∞ equals*

$$\sum_{\mathfrak{a}} \sum_{\nu < 0} a_{\mathfrak{a}}(\nu) \sum_c \frac{2\pi}{c} \sqrt{\frac{|\nu|}{x}} S_{\mathfrak{a}\infty}(\nu, x; c) I_1\left(\frac{4\pi}{c} \sqrt{|\nu|x}\right).$$

Here $\mathfrak{a}$ runs over inequivalent cusps, ν over the exponents of the principal part at $\mathfrak{a}$ (in the local parameter of width $w_{\mathfrak{a}}$, normalised so that $\nu \in w_{\mathfrak{a}}^{-1}\mathbb{Z} + \kappa_{\mathfrak{a}}$), $a_{\mathfrak{a}}(\nu)$ are the principal-part coefficients, c runs over the allowed moduli for the pair $(\mathfrak{a}, \infty)$, and $S_{\mathfrak{a}\infty}$ is the cusp-pair Kloosterman sum with the character.

Apply Theorem 2.2 to F^* with $x = 24n - m(t-1)$, $n > \mu$. Each triple $(\mathfrak{a}, \nu, c)$ gives a term whose Bessel factor depends only on the rate $\beta = \sqrt{|\nu|}/c$, up to the normalisation $x = 24(n - \mu)$.

2.2. Growth rates and groups.

Definition 2.3. For a term T with rate β_T put

$$B_{\beta}(n) := \frac{\beta}{\sqrt{n - \mu}} I_1(4\pi\beta\sqrt{24(n - \mu)}).$$

The *group amplitude* is

$$G_{\beta}(n) := \sum_{T: \beta_T = \beta} (\text{constant}) \cdot S_{\mathfrak{a}_T\infty}(\nu_T, x; c_T),$$

so that $c(n) = \sum_{\beta} G_{\beta}(n) B_{\beta}(n)$ for $n > \mu$.

Only finitely many $(\mathfrak{a}, \nu)$ occur, since the principal parts are finite. The Weil bound $|S_{\mathfrak{a}\infty}(\nu, x; c)| \ll_{\epsilon, N, \nu} c^{1/2+\epsilon} \gcd(x, c)^{1/2}$, together with $I_1(y) \ll y$ for small y , makes the series absolutely convergent for weight 0. For each $\epsilon > 0$ only finitely many c satisfy $\sqrt{|\nu|}/c > \epsilon$. Each G_{β} is periodic in n , with period dividing lcm of the c_T in the group.

Lemma 2.4 (uniqueness). *Let (G_{β}) and (G'_{β}) be two families of periodic functions indexed by a set of rates $\beta > 0$ that has no accumulation point other than 0. Suppose*

$$\sum_{\beta} G_{\beta}(n) B_{\beta}(n) = \sum_{\beta} G'_{\beta}(n) B_{\beta}(n) \quad \text{for all } n > \mu,$$

and that both series converge absolutely, with $\sum_{\beta \leq \epsilon} |G_{\beta}(n)| B_{\beta}(n) = O(e^{4\pi\epsilon\sqrt{24n} + o(\sqrt{n})})$ for each ϵ . Then $G_{\beta} = G'_{\beta}$ for every β .

Proof. Put $D_{\beta} = G_{\beta} - G'_{\beta}$, and suppose some $D_{\beta} \neq 0$. Let β_0 be the largest such rate; it exists because the rates above any positive level are finite in number. Let P be a period of D_{β_0} , and choose a class $r \bmod P$ with $D_{\beta_0}(r) \neq 0$.

For $n \equiv r \pmod{P}$, subtracting the two expansions gives

$$0 = D_{\beta_0}(r) B_{\beta_0}(n) + \sum_{\beta < \beta_0} D_{\beta}(n) B_{\beta}(n).$$

Let $\beta_1 < \beta_0$ be the next rate. The tail is $O(e^{4\pi\beta_1\sqrt{24n} + o(\sqrt{n})})$, while $|B_{\beta_0}(n)| \gg n^{-3/4} e^{4\pi\beta_0\sqrt{24n}}$. This is a contradiction as $n \rightarrow \infty$. $\square$

The same argument, applied to the expansion of $c(n)$ restricted to a residue class, yields the principle used throughout.

Corollary 2.5. *Fix a class $C = \{n \equiv r \pmod{L}\}$. Suppose that for some β_0 we have $G_\beta(n) = 0$ for all $\beta > \beta_0$ and all $n \in C$, while G_{β_0} is a nonzero constant g on C . Then $\operatorname{sgn} c(n) = \operatorname{sgn} g$ for all sufficiently large $n \in C$. If instead $G_\beta \equiv 0$ on C for every β , then $c(n) = 0$ for all $n \in C$ with $n > \mu$.*

Note that G_{β_0} is real on C , since $c(n)$ is real and Lemma 2.4 applies to the expansion and its complex conjugate.

3. THE LEAST PERIOD IS DIVISIBLE BY t

The largest rate comes from the cusps of $\Gamma_0(t)$ of denominator exactly $k = t$, which are $\Gamma_0(t)$ -equivalent to ∞ for F . On that group the principal part is $q^{-\mu}$, and by [10, (2.17)] the group amplitude, suitably normalised, is Wang's

$$\alpha(n) = \sum_{h \bmod t} {}^*\psi(h) e(-hn/t), \quad |\psi(h)| = 1.$$

Call a class $r \bmod t$ *non-degenerate* if $\alpha(r) \neq 0$. By Corollary 2.5, for $n \equiv r \pmod{t}$ in a non-degenerate class, $\operatorname{sgn} c(n) = \operatorname{sgn} \alpha(r)$ eventually.

Proposition 3.1. *If $(c(n))$ is UPS, then its least period of sign is divisible by t .*

Proof. Let g be a proper divisor of t and s any residue. Then

$$\sum_{j \bmod t/g} \alpha(s + jg) = \sum_h {}^*\psi(h) e(-hs/t) \sum_j e(-h jg/t) = 0,$$

because $t/g \nmid h$ for every unit h . Since the $\psi(h)$ are nonzero, $\alpha \not\equiv 0$. So there is a coset $s + g\mathbb{Z}/t\mathbb{Z}$ on which α is not identically zero. On that coset α takes both a positive and a negative value, at non-degenerate classes.

Now suppose the least period L satisfies $t \nmid L$, and put $g = \gcd(L, t) < t$. The multiples of L reduce mod t to exactly $g\mathbb{Z}/t\mathbb{Z}$. So for every non-degenerate r and every j , the classes r and $r + jg$ have the same eventual sign. That is impossible on the coset just found. $\square$

4. PRIME LEVEL

Throughout this section $t = p$ is prime. The cusps of $\Gamma_0(p)$ are ∞ and 0. F has its only pole at ∞ , with principal part $\sum_{0 \leq e < \mu} c(e)q^{e-\mu}$, and it is holomorphic at 0.

4.1. The multiplier at prime level. Let $M = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in \Gamma_0(p)$ with $c > 0$, and put $M_p = \begin{pmatrix} a & pb \\ c/p & d \end{pmatrix}$. Since $\eta(pM\tau) = \eta(M_p(p\tau))$, the multiplier of F is

$$(2) \quad \nu_F(M) = (\nu_\eta(M)/\nu_\eta(M_p))^m.$$

Knopp's formula [7, Ch. 4, Thm. 2] gives, for $c > 0$,

$$\nu_\eta(M) = \left(\frac{d}{c}\right) e\left(\frac{(a+d)c - bd(c^2 - 1) - 3c}{24}\right) \quad (c \text{ odd}), \quad \nu_\eta(M) = \left(\frac{c}{d}\right) e\left(\frac{(a+d)c - bd(c^2 - 1) + 3d - 3 - 3cd}{24}\right) \quad (c \text{ even}).$$

Lemma 4.1. *Let $p \geq 5$ and $c = p^\alpha c'$ with $\alpha \geq 1$ and $p \nmid c'$. Then $\nu_F(M) = \chi_p(d)^m \rho(M)$, where $\rho(M) = e(R/24c)$ for an integer R with $p^\alpha \mid R$.*

Proof. Characters. If c is odd, then $\left(\frac{d}{c}\right)\left(\frac{d}{c/p}\right)^{-1} = \chi_p(d)$. If c is even, then $\left(\frac{c}{d}\right)\left(\frac{c/p}{d}\right)^{-1} = \left(\frac{p}{d}\right)$, and quadratic reciprocity gives $\left(\frac{p}{d}\right) = \chi_p(d)\epsilon(d)$ with ϵ a character modulo 4.

Additive parts. Put $j = c/p$. The difference of the additive exponents is

$$\frac{p-1}{24} [(a+d)j - bd(pj^2 + 1) - 3j],$$

plus $-3dj(p-1)/24$ when c is even. Multiply by $24c$. The resulting integer is a combination of $c^2(a+d)/p$, $cbd(pj^2+1)$, $3cj$ and $3dc^2/p$. Each of these is divisible by $p^{2\alpha-1}$ or by c , hence by p^α . $\square$

4.2. Amplitudes. Terms of the exact formula for F on $\Gamma_0(p)$ are indexed by $e \in [0, \mu]$ and by $c \equiv 0 \pmod{p}$. Their amplitudes are

$$(3) \quad \mathcal{A}_{c,e}(n) = c(e) \sum_{d \bmod c} {}^*\overline{\nu_F(M_d)} e\left(\frac{(e-\mu)\bar{d} + (n-\mu)d}{c}\right),$$

where the fractional exponents are combined into a single exponential modulo $24c$. By Lemma 4.1 and the Chinese remainder theorem (valid because $p \geq 5$), for $c = p^\alpha c'$,

$$(4) \quad \mathcal{A}_{c,e}(n) = c(e) T_\alpha(u, v) R_{c',e}(n), \quad T_\alpha(u, v) = \sum_{d \bmod p^\alpha} {}^*\chi_p(d)^m e\left(\frac{ud + v\bar{d}}{p^\alpha}\right),$$

where

$$u \equiv (n - \mu) \bar{c}', \quad v \equiv (e - \mu) \bar{c}' \pmod{p^\alpha},$$

reading $n - \mu$ and $e - \mu$ in $\mathbb{Z}_{(p)}$. In particular $\chi_p(uv) = \chi_p((n - \mu)(e - \mu))$ does not depend on c' . For $c = p$ we have T_1 , and the leading amplitude is $\alpha(n) = \mathcal{A}_{p,0}(n)$ up to a unit.

Lemma 4.2. *Let p be an odd prime.*

- (a) *For m even, $T_1(u, v)$ is the Kloosterman sum $S(u, v; p)$, and it is never 0.*
- (b) *Let m be odd and $p \nmid uv$. Then $T_\alpha(u, v) = 0$ for every $\alpha \geq 1$ if $\chi_p(uv) = -1$, and $T_\alpha(u, v) \neq 0$ if $\chi_p(uv) = 1$.*
- (c) *For m odd, $T_1(u, 0) = \chi_p(u) \mathfrak{g}_p \neq 0$ if $p \nmid u$, and $T_1(0, v) = \chi_p(v) \mathfrak{g}_p \neq 0$ if $p \nmid v$. Here $\mathfrak{g}_p$ is the quadratic Gauss sum. Moreover $T_1(0, 0) = 0$.*

Proof. (a) In $\mathbb{Z}[\zeta_p]$ each term is $\equiv 1 \pmod{1 - \zeta_p}$, so $S \equiv p - 1 \equiv -1$.

(b) For $\alpha = 1$, Salie's evaluation gives $T_1(u, v) = \chi_p(v) \mathfrak{g}_p \sum_{x^2 \equiv 4uv} e(x/p)$. If $\chi_p(uv) = -1$ the sum is empty. If $\chi_p(uv) = 1$ it has the two terms $x = \pm x^*$ with $x^{*2} \equiv 4uv$, giving $2\chi_p(v) \mathfrak{g}_p \cos(2\pi x^*/p)$; since $p \nmid x^*$ and $p \geq 5$, the cosine is nonzero, so $T_1 \neq 0$. For $\alpha \geq 2$, put $\beta = \lfloor \alpha/2 \rfloor \geq 1$ and write $d = d_0(1 + p^{\alpha-\beta}y)$ with $y \bmod p^\beta$. Then $\chi_p(d) = \chi_p(d_0)$, and $\bar{d} \equiv \bar{d}_0(1 - p^{\alpha-\beta}y) \pmod{p^\alpha}$. Summing over y gives p^β times the indicator of $ud_0 \equiv v\bar{d}_0 \pmod{p^\beta}$, that is, of $ud_0^2 \equiv v$. This forces $\chi_p(uv) = 1$, so for $\chi_p(uv) = -1$ the sum is 0. For $\chi_p(uv) = 1$ the surviving d_0 are the two square roots, and the argument of part (b) at level α again leaves a nonzero cosine, so $T_\alpha \neq 0$.

(c) These are Gauss sums; $T_1(0, 0) = \sum \chi_p(d) = 0$. $\square$

Lemma 4.3 (multiplier at $c = p$). *For $p \geq 5$,*

$$e^{-\pi i s(h,p)} = C_p \chi_p(h) e\left(-\frac{24(h + \bar{h})}{p}\right), \quad |C_p| = 1.$$

Consequently, with $u \equiv \mu - r$ and $v \equiv \mu - e$ taken mod p ,

$$\alpha(r) = C_p^m T_1(u, v) \quad \text{at } e = 0.$$

Proof. Take $c = p$ in Knopp's formula, using $e^{\pi i s(d,c)} = e((a+d)/24c)/\varepsilon(a, b, c, d)$ from [2, Thm. 3.4]. Then reduce the exponent modulo p using $ad \equiv 1$ and the fact that 24 is invertible mod p . $\square$

Proof of Theorem 1.3. (a), $p \geq 5$ and m even. By Lemmas 4.3 and 4.2(a), $\alpha(r) \neq 0$ for all r . The $c = p$, $e = 0$ group has the unique largest rate $\sqrt{\mu}/p$ and constant amplitude on each class mod p , so Corollary 2.5 gives $\text{sgn } c(n) = \text{sgn } \alpha(n \bmod p)$ eventually. By Proposition 3.1 the least period is p .

(a), $p = 3$. Here $s(1, 3) = 1/18 = -s(2, 3)$, so $\alpha(r) = 2 \cos(\pi m/18 + 2\pi r/3)$. This vanishes if and only if $m \equiv 3 \pmod{6}$, and otherwise the same argument applies.

(b). Let r be degenerate, and let e_0 be the least $e < \mu$ with $c(e) \neq 0$ and $T_1(\mu-r, \mu-e) \neq 0$, if one exists. By (4) and Lemma 4.2(b), every term (c, e) with e “bad” (that is, $\chi_p((\mu-r)(\mu-e)) = -1$) vanishes on the class at every level c . Terms with $c(e) = 0$ vanish trivially. Hence all terms of rate larger than $\sqrt{\mu - e_0}/p$ vanish on the class, while the (p, e_0) term is a nonzero constant there; Corollary 2.5 applies. If no such e_0 exists, every term vanishes on the class, and $c(n) = 0$ there.

In the exceptional case $r \equiv \mu \equiv 0 \pmod{p}$ we have $p \mid m$, and $e = 1$ works, because $c(1) = -m \neq 0$ and $T_1(0, -1) \neq 0$ by Lemma 4.2(c). For $p = 3$ and $m \equiv 3 \pmod{6}$, the order-1 amplitude is $2c(1) \cos(-\pi m/18 + 2\pi(2-r)/3)$, which is nonzero on the degenerate class. Its rate $\sqrt{\mu - 1}/3$ exceeds that of every term with $k \geq 6$ as soon as $\mu > 4/3$, i.e. for $m \geq 21$. For $m = 15$ the only faster term, $(k, e) = (6, 0)$, vanishes identically on the degenerate class. This is a finite exact computation in $\mathbb{Q}(\zeta_{144})$, confirmed numerically to 10^{-29} . The cases $m = 3, 9$ are due to Wang. $\square$

5. COMPOSITE LEVEL: THE SCALING LAW

Let S be the set of primes dividing $N = 576t$ or dividing the numerator or denominator of $|\nu|$ for some principal-part exponent ν of F^* at some cusp. This set is finite.

Lemma 5.1. *If two terms T, T' have the same rate, then $c_T/c_{T'}$ is an S -unit.*

Proof. $c_T/c_{T'} = \sqrt{|\nu_T|/|\nu_{T'}|}$, which is rational, and its prime factors divide numerators or denominators of the $|\nu|$. $\square$

5.1. The Galois scaling law. We work with the $\Gamma_0(t)$ expansion of F . For a term $T = (k, \gamma)$ of cusp type $d = \gcd(k, t)$ its amplitude is

$$(5) \quad A_T(n) = \sum_{h \bmod k} {}^*s_T(h; n), \quad s_T(h; n) = \left(\frac{t}{d}\right)^{m/2} (\omega_{h,k}^{-1} \omega_{th/d, k/d})^m \kappa_T(h) e\left(-\frac{hn}{k}\right),$$

where $\kappa_T(h)$ is the coefficient of order γ in the expansion of the transformed product. All monomials of order γ carry the same phase $e(\Lambda_T h'/k)$, so $\kappa_T(h) = \kappa_T e(\Lambda_T h'/k)$ with $hh' \equiv -1 \pmod{k}$.

Every $s_T(h; n)$ lies in $\mathbb{Q}(\zeta_M, \sqrt{t/d})$ with $M = 24k$. For an integer g prime to $6tk$ we fix an automorphism σ_g of $\overline{\mathbb{Q}}$ by two requirements: $\sigma_g(\zeta_M) = \zeta_M^g$, and $\sigma_{\bar{q}}(\sqrt{D}) = (\frac{D^*}{q})\sqrt{D}$ for every squarefree $D > 1$ dividing t , where D^* is the discriminant of $\mathbb{Q}(\sqrt{D})$ and $q\bar{q} \equiv 1 \pmod{M}$. (Note the symbol is evaluated at q , not at $\bar{q}$. The two agree when $D^* \mid M$, since then $q\bar{q} \equiv 1 \pmod{D^*}$; when $D^* \nmid M$ they need not agree, and it is the value at q that is required below.)

Such a σ_g exists. Write $k = dK$ with $d = \gcd(k, t)$ and $\gcd(K, t/d) = 1$; the primes of D divide t/d and are therefore prime to K , so $D^* \mid M = 24k$ holds if and only if $D^* \mid 24d$. This can fail — for instance $t = 10$, $d = 2$, $D = 5$ — and then $\sqrt{D} \notin \mathbb{Q}(\zeta_M)$, so the action of an automorphism on $\sqrt{D}$ is not determined by its action on ζ_M and must be prescribed. When $D^* \mid M$ the prescribed value coincides with the one forced by $\sigma_{\bar{q}}(\zeta_{D^*}) = \zeta_{D^*}^{\bar{q}}$ through the Gauss sum $\sqrt{D^*} = \sum_a (\frac{D^*}{a}) \zeta_{D^*}^a$, which gives $(\frac{D^*}{q}) = (\frac{D^*}{\bar{q}})$, so the two requirements are compatible; when $D^* \nmid M$ the fields $\mathbb{Q}(\zeta_M)$ and $\mathbb{Q}(\sqrt{D})$ are linearly disjoint and the value may be prescribed freely. The prescriptions for different D are mutually consistent because $\sqrt{D_1}\sqrt{D_2}$ is a rational multiple of $\sqrt{D_3}$, D_3 the squarefree part of $D_1 D_2$, and the Kronecker symbol is multiplicative: $(\frac{D_1^*}{g})(\frac{D_2^*}{g}) = (\frac{D_3^*}{g})$. Finitely many quadratic fields occur, one for each divisor d of t , so a single σ_g serves all terms at once. With this convention the sign in Step 3 below is the prescribed value, not a quantity computed from $\sigma_g(\zeta_M)$. By Lemma 5.1, the moduli within one equal-growth group differ only by primes of S .

Lemma 5.2 (summand factorisation). *Let $q \notin S$ be prime and $T = (k, \gamma)$. For every h prime to $24kq$,*

$$s_{(kq, \gamma)}(h; n) = \sigma_{\bar{q}}(s_{(k, \gamma)}(h; n)) \cdot e\left(\frac{\bar{k}((\mu - n)h + \gamma\bar{h})}{q}\right),$$

where $\bar{q}$ is inverse to q modulo $24k$, and $\bar{k}, \bar{h}$ are inverses modulo q . There is no further constant.

Proof. Throughout, h is prime to $24kq$, and $\bar{h}$ denotes its inverse modulo $24kq$, hence also modulo $24k$. We use the explicit formulas, valid for $\gcd(h, 24k) = 1$:

$$(O) \quad \omega_{h,k} = \left(\frac{h}{k}\right) i^{\frac{k-1}{2}} e\left(\frac{(1-k^2)(h+\bar{h})}{24k}\right) \quad (k \text{ odd}), \quad (E) \quad \omega_{h,k} = \left(\frac{k}{h}\right) e\left(\frac{\bar{h}(1-k^2) + h(2k^2 - 3k + 1)}{24k}\right) \quad (k \text{ even}).$$

These follow from Apostol [2, Thm. 3.4] (the formula $e^{\pi is(d,c)} = e((a+d)/24c)/\varepsilon(a, b, c, d)$ with Apostol's evaluation of ε), after choosing $a \equiv \bar{h} \pmod{24k}$. They were checked on 3409 pairs.

Step 1: $\omega_{h,kq}$ versus $\omega_{h,k}$. Write $e(X/24kq) = e(X\bar{q}/24k) e(X\bar{24k}/q)$, where the first inverse is modulo $24k$ and the second modulo q .

(O) Here $(h/kq) = (h/k)(h/q)$. In the part modulo $24k$ we use $1 - k^2q^2 \equiv 1 - k^2 \pmod{24k}$, which holds because $24 \mid k(q^2 - 1)$. So that part equals $\sigma_{\bar{q}}$ of the level- k exponential. The i -powers give

$$i^{(kq-1)/2} / \sigma_{\bar{q}}(i^{(k-1)/2}) = i^{[(kq-1)-q(k-1)]/2} = i^{(q-1)/2},$$

using $\bar{q} \equiv q \pmod{8}$.

(E) Here $(kq/h) = (k/h)(q/h) = (k/h)(h/q)\chi_{-4}(h)^{(q-1)/2}$ by reciprocity. In the part modulo $24k$, the $\bar{h}$ -coefficient is again $\bar{q}(1 - k^2)$. The h -coefficient is

$$\bar{q}(2k^2 - 3k + 1) + \bar{q}[2k^2(q^2 - 1) - 3k(q - 1)].$$

The term $2k^2(q^2 - 1)$ is $\equiv 0 \pmod{24k}$. The term $-3k(q - 1)$ contributes $e(-(q - 1)\bar{q}h/8)$. For odd h one checks:

- if $q \equiv 1 \pmod{4}$, this factor equals $(-1)^{(q-1)/4}$, and $\chi_{-4}(h)^{(q-1)/2} = 1$;
- if $q \equiv 3 \pmod{4}$, writing $a = \frac{q-1}{2}\bar{q}$ (odd), it equals $i^{-ah} = i^{-a}\chi_{-4}(h)$, which cancels $\chi_{-4}(h)^{(q-1)/2} = \chi_{-4}(h)$ and leaves i^{-a} .

In both cases the leftover constant equals $i^{(q-1)/2}$, because $(q - 1)(q + 1)/2 \equiv 0 \pmod{4}$.

Hence, in all cases,

$$(6) \quad \omega_{h,kq} = \sigma_{\bar{q}}(\omega_{h,k}) \cdot \left(\frac{h}{q}\right) i^{\frac{q-1}{2}} e\left(\frac{\bar{24k}(h + \bar{h})}{q}\right).$$

For the part modulo q we used $1 - k^2q^2 \equiv 1$ and $2k^2q^2 - 3kq + 1 \equiv 1 \pmod{q}$.

Step 2: the second factor. Let $K = k/d$ and lift th/d to $H_0 = (t/d)h + K\ell$ with $\gcd(H_0, 6) = 1$. Since $\gcd(K, t/d) = 1$, ℓ may be taken independent of h , and even whenever $2 \nmid t/d$. At level kq use $H = (t/d)h + Kq\ell'$ with $q\ell' \equiv \ell \pmod{24}$. Then $H \equiv H_0 \pmod{24K}$ and $H \equiv (t/d)h \pmod{q}$. Applying (6) to (H, K) gives

$$\omega_{H,Kq} = \sigma_{\bar{q}}(\omega_{H_0,K}) \left(\frac{(t/d)h}{q}\right) i^{\frac{q-1}{2}} e\left(\frac{\bar{24K}(H + \bar{H})}{q}\right).$$

Step 3: combine. In $\omega_{h,kq}^{-m} \omega_{H,Kq}^m$ the factors $i^{\pm m(q-1)/2}$ cancel, and the characters leave $(\frac{t/d}{q})^m$. For odd m , the Galois sign of $(t/d)^{m/2}$ is $\sigma_{\bar{q}}(\sqrt{D})/\sqrt{D} = (\frac{D^*}{q})$ by the convention fixed above, where D is the squarefree part of t/d and D^* its discriminant. Since $(\frac{D^*}{q}) = (\frac{t/d}{q})$, the two signs cancel.

The phase $e(\Lambda_T h'/kq)$ and the term $e(-hn/kq)$ split in the same way. Their parts modulo $24k$ are $\sigma_{\bar{q}}$ of the level- k phases, and their parts modulo q are $e(-\bar{k}\Lambda_T \bar{h}/q)$ and $e(-\bar{k}nh/q)$. The rational coefficient κ_T is fixed by $\sigma_{\bar{q}}$.

The part modulo q collects to

$$\bar{k} \left[\frac{m}{24} \left(- (h + \bar{h}) + (th + \frac{d^2}{t} \bar{h}) \right) - nh - \Lambda_T \bar{h} \right] = \bar{k} [(\mu - n)h + \gamma \bar{h}],$$

using $\gamma = \frac{m(d^2 - t)}{24t} - \Lambda_T$. The identity was confirmed numerically: the constant equals 1 for every member of the 26 multi-member groups tested (`eps_table.py`). $\square$

Proposition 5.3 (Galois scaling law). *Let $q \notin S$ be prime and G_β an equal-growth group. Then*

$$G_{\beta/q}(n) = S(\mu - n, \beta^2; q) \sigma_{\bar{q}}(G_\beta(n)),$$

where $S(\cdot, \cdot; q)$ is the Kloosterman sum, with $\beta^2 = \gamma/k^2$ read modulo q . The Kloosterman factor is never zero. In particular $G_{\beta/q}(n) = 0$ if and only if $G_\beta(n) = 0$.

Proof. By Lemma 5.1 the members of $G_{\beta/q}$ are exactly the members of G_β with moduli multiplied by q . Sum Lemma 5.2 over $h \bmod kq$, using $h \leftrightarrow (h \bmod 24k, h \bmod q)$. The q -sum is

$$\sum_{h \bmod q} {}^*e(\bar{k}((\mu - n)h + \gamma \bar{h})/q) = S(\mu - n, \bar{k}^2 \gamma; q),$$

and $\bar{k}^2 \gamma \equiv \gamma/k^2 = \beta^2$ is common to all members. A Kloosterman sum modulo a prime is $\equiv -1 \pmod{1 - \zeta_q}$ (or is a nonzero Ramanujan sum), hence nonzero. Galois automorphisms preserve vanishing. $\square$

Proposition 5.4 (stable range). *Let $q \in S$. Put $e_q = \kappa_q = 3$ if $q = 2$ and $e_q = \kappa_q = 1$ if q is odd, let*

$$W_q = v_q(24t) + \max_{\gamma} v_q(\text{numerator of } 24t\gamma),$$

the maximum being over all pole orders, and set

$$A_q = \max \{v_q(24t) + v_q(t) + 3, 2(W_q + e_q) + \kappa_q + 3\}.$$

Let G_β be a group all of whose moduli satisfy $v_q(k) \geq A_q + 1$, and let $G_{\beta q}$ be the group obtained by dividing every modulus by q . Then for every n ,

$$G_\beta(n) \neq 0 \implies G_{\beta q}(n) \neq 0.$$

Proof. Write $k = q^a k_0$ with $q \nmid k_0$, and put $A = a + v_q(24)$. Take $a \geq v_q(24t) + v_q(t) + 3$. Then scaling by q does not change the cusp type d , and moreover:

- $1 - k^2 \equiv 1 \pmod{q^A}$;
- $2k^2 - 3k + 1 \equiv 1 - 3k \pmod{q^A}$, which matters only for $q = 2$, where it contributes the factor $e(-h/8)$;
- the bounded-denominator phases of the multiplier are trivial on the q -part.

Factorisation. By the Chinese remainder theorem each summand splits into a prime-to- q part and a q -part. By Steps 1–3 of Lemma 5.2, applied to the prime-to- q modulus (where the needed congruence holds because the prime-to- q part of 24 divides $q^2 - 1$), the prime-to- q part at level k/q is σ'^{-1} of the prime-to- q part at level k . Here σ' acts as $\sigma_{\bar{q}}$ on roots of unity of order prime to q and trivially on those of q -power order.

The q -part of the member $T_i = (k_i, \gamma_i)$ is computed exactly as the part modulo q in Lemma 5.2. After the substitution $h \mapsto k'_i h$, where k'_i is the prime-to- q part of $24k_i$, it equals

$$\psi(k'_i) \mathcal{T}^{(A_i)}(x, y_i), \quad \mathcal{T}^{(A)}(x, y) = \sum_{h \bmod q^A} {}^*\psi(h) \theta(h) e((xh + y\bar{h})/q^A).$$

Here:

- ψ is a real character of conductor dividing 8 (if $q = 2$) or q (if q is odd), determined by t, m and d ;

- $\theta(h) = e(-ch/8)$ for $q = 2$ and $\theta = 1$ otherwise;
- $x = c_1(\mu - n)$ with c_1 a fixed unit;
- $y_i = c_2\gamma_i k_i'^{-2} = c_3\beta^2 q^{2a_i}$ with c_2, c_3 fixed units.

In particular y_i depends only on β and a_i .

Subgroups. Split G_β into subgroups $G_\beta^{(\delta)}$ according to the q -exponent $a_i = a + \delta$. Within one subgroup, the q -parts are $\psi(k'_i) \mathcal{T}^{(A+\delta)}(x, y_\delta)$ with a common y_δ , so

$$G_\beta^{(\delta)}(n) = \mathcal{T}^{(A+\delta)}(x, y_\delta) \cdot \sum_i \psi(k'_i) P_i(n),$$

where the P_i are the prime-to- q parts. Likewise

$$G_{\beta q}^{(\delta)}(n) = \mathcal{T}^{(A+\delta-1)}(x, y_\delta) \cdot \sigma'^{-1} \left(\sum_i \psi(k'_i) P_i(n) \right),$$

because $\psi(k'_i) = \pm 1$ is fixed by σ' .

Stationary phase. Let $B \geq 2(v_q(y) + e_q) + \kappa_q + 2$; this holds at every level used, since $v_q(y_\delta) \leq W_q$. Put $\beta' = \lfloor B/2 \rfloor$. Then $B - \beta' \geq \kappa_q$, so ψ and θ are invariant under the substitution below, and $\beta' > v_q(y) + e_q$. Substitute $h = h_0(1 + q^{B-\beta'}z)$. This leaves ψ and θ invariant, and summing over z gives

$$\mathcal{T}^{(B)}(x, y) = q^{\beta'} \sum_{xh_0^2 \equiv y \pmod{q^{\beta'}}} \psi(h_0) \theta(h_0) e((xh_0 + y\bar{h}_0)/q^B).$$

- (i) *Solvability.* The congruence has a solution if and only if $v_q(x) = v_q(y)$ and $(x/q^v)(y/q^v)$ is a square modulo q (modulo 8 if $q = 2$), where $v = v_q(y)$. This condition does not depend on B . Since $v_q(y_\delta) = v_q(c_3\beta^2) + 2(a + \delta)$ is different for different δ , at most one subgroup has a solvable congruence at a given n . All other subgroups vanish at n , at both levels.
- (ii) *Non-vanishing.* Suppose the congruence is solvable. For odd q the solutions are $h_0 \equiv \pm h^*$. The sum equals $q^{B/2}$, times a nonzero Gauss-sum factor when B is odd, times

$$\psi(h^*) (e(2x^*/q^B) + \psi(-1)e(-2x^*/q^B)), \quad x^{*2} \equiv xy.$$

This is $2 \cos$ or $2i \sin$ of $4\pi x^*/q^B$. Since $v_q(x^*) = v$ is fixed and B is large, it is nonzero.

For $q = 2$ the solutions are $\pm h^*$ and $\pm h^*(1 + 2^{B-1})$. Since $B - 1 \geq 3$, the factors ψ and θ take the same values at h^* and $h^*(1 + 2^{B-1})$. Also $2x^*(1 + 2^{B-1})/2^B \equiv 2x^*/2^B \pmod{1}$ because x^* is an integer. So the four terms pair up to

$$2\psi(h^*) \left(\theta(h^*)e(2x^*/2^B) + \psi(-1)\theta(-h^*)e(-2x^*/2^B) \right).$$

This vanishes only if $e(h^*c/4 - 4x^*/2^B) = -\psi(-1)$, that is only if $4x^*/2^B \equiv h^*c/4 + \{0, \frac{1}{2}\} \pmod{1}$. That forces $v_2(x^*) \leq B - 4$ to equal a specific value. It is impossible once $B > v + 4$, because $v_2(x^*) = v$ is fixed.

Conclusion. If $G_\beta(n) \neq 0$, then by (i) exactly one subgroup $G_\beta^{(\delta)}$ is nonzero at n . Its factor $\mathcal{T}^{(A+\delta)}(x, y_\delta)$ is nonzero and $\sum_i \psi(k'_i) P_i(n) \neq 0$. By (i) and (ii) at level $A + \delta - 1 \geq A_q$, $\mathcal{T}^{(A+\delta-1)}(x, y_\delta) \neq 0$. Hence $G_{\beta q}^{(\delta)}(n) \neq 0$, while all other subgroups of $G_{\beta q}$ vanish at n . So $G_{\beta q}(n) \neq 0$. $\square$

6. FINITENESS AND THE PROOF OF THEOREM 1.2

Definition 6.1. A modulus c is *rigid* if every prime factor of c lies in S and $v_q(c) \leq A_q + 1 + 2\Delta_q$ for all $q \in S$. A group is rigid if all its moduli are rigid. Put

$$L = \text{lcm}(t, \text{all rigid moduli}),$$

which is finite.

Lemma 6.2. *Every non-rigid group is obtained from a rigid group G_{β_0} by finitely many scalings: first by primes $q \in S$ in the stable range, then by an integer c' all of whose prime factors lie outside S . Moreover,*

$$G_{\beta_0/(c'c)}(n) = \left(\prod \mathcal{K} \right) \sigma(G_{\beta_0}(n)),$$

where σ is a Galois automorphism and $\prod \mathcal{K}$ is a product of scalars determined by n and the scalings.

Proof. Remove from the moduli the primes outside S ; by Lemma 5.1 they are common to all members. Then lower the q -exponents to the cap, which is possible by Lemma 5.1 and the choice of $2\Delta_q$. Apply Proposition 5.3 at each step. $\square$

Lemma 6.3. *If $\beta_1/c_1 = \beta_2/c_2$ with G_{β_1}, G_{β_2} rigid and c_1, c_2 composed of primes outside S , then $c_1 = c_2$ and $\beta_1 = \beta_2$.*

Proof. β_1/β_2 is an S -unit by Lemma 5.1, and c_1/c_2 is a unit away from S . Their equality forces both ratios to be 1. $\square$

Proof of Theorem 1.2. Let G_β be any group with $G_\beta(n) \neq 0$ for some n . By Lemma 6.2, G_β is obtained from a rigid group G_{β_0} , with $\beta_0 \geq \beta$, by scalings. By Proposition 5.3 (for primes outside S , where the amplitude is multiplied by a nonzero Kloosterman sum and a Galois automorphism is applied, without changing n) and Proposition 5.4 (for primes of S beyond the caps), $G_\beta(n) \neq 0$ implies $G_{\beta_0}(n) \neq 0$.

Consequently, at every n the first non-vanishing group (in order of decreasing rate) is rigid. Rigid group amplitudes are constant on classes modulo L . Fix a class C modulo L .

- If some rigid group is nonzero on C , let G^* be the fastest one. Every faster group, rigid or not, vanishes on C : rigid ones by the choice of G^* , and non-rigid ones because their rigid ancestor is faster than G^* and hence vanishes on C . Moreover G^* is a nonzero real constant on C , since $c(n)$ is real. Corollary 2.5 gives $\text{sgn } c(n) = \text{sgn } G^*$ for large $n \in C$.
 - If every rigid group vanishes on C , then every group vanishes on C , and $c(n) = 0$ there.
- Hence $(c(n))$ is UPS with period dividing L . Proposition 3.1 gives $t \mid$ least period. $\square$

7. NUMERICAL VERIFICATION

All steps were checked numerically; the scripts accompany the paper.

- The eta multiplier formula was checked on 39 random matrices, and the resulting explicit Dedekind-sum phases on 3409 pairs.
- Twisted Salié sums modulo p^α were checked in 408 cases.
- The prime-level vanishing criterion was checked in 486 cases at levels up to p^2 and $5p^2$. The predicted degenerate classes agree with exact computation in all 5605 cases.
- The Galois scaling law (Proposition 5.3) holds in 364 group tests, including $t = 6, 8, 10, 12, 15$. Its summand-level form (Lemma 5.2) holds in 20 cases, and its stable-range form (Proposition 5.4) in 112 cases.
- A regression of the resulting sign predictions against exact coefficients covers $t \leq 12$ and $m(t-1) \leq 60$ (21 899 coefficients), with no discrepancy once the truncation is large enough.
- *Stress tests.*
 - The exact identity of Lemma 5.2 holds on 45 randomly chosen terms ($t \leq 20$, $m \leq 12$, both parities of k).
 - Proposition 5.4 holds in 26 checks with n chosen to vary $v_q(24n - m(t-1))$. The local lemma underlying it (vanishing of $\mathcal{T}^{(B)}(x, y)$ exactly when the congruence is unsolvable, at the explicit threshold $B \geq 2(v_q(y) + e_q) + \kappa_q + 2$) holds in 186 direct tests. These include $q = 2$ with all four characters modulo 8 and all eight twists θ , up to $B = 13$.

- *End to end.* Predicting every sign from *rigid* groups alone, as the proof asserts is possible, reproduces brute force exactly: 15 613 coefficients for 13 pairs, including Wang’s exceptional cases (3, 9), (4, 4) and (5, 5).
- Brute-force least sign periods on $[N/3, N]$ with $N = 2400$ equal t for 12 pairs beyond Wang’s range. These windows include n with $v_q(24n - m(t - 1))$ up to 8.
- Brute-force sign periods were computed for all 124 pairs with $t \leq 12$ and $24 < m(t - 1) \leq 72$. The least period is exactly t in every case.
- *Beyond the tabulated range.* The least sign period was also computed for 22 pairs with $14 \leq t \leq 27$ and $24 < m(t - 1) \leq 72$, including $t = 16, 18, 20, 24, 25, 27$ (square factors) and $t = 15, 21, 22$ (several prime divisors). In every case the least period is exactly t . These examples show the eventual threshold growing with t : the sequence is t -periodic from $n = 799$ for $(t, m) = (18, 3)$, from $n = 1170$ for $(20, 2)$, from $n = 1351$ for $(22, 2)$, from $n = 1491$ for $(27, 2)$, and from $n = 2082$ for $(24, 2)$. The last of these lies beyond the range $N = 2400$ used for the $t \leq 12$ windows above, and a window beginning before the threshold shows no period at all; this is the expected behaviour, not a failure.

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